# Supplementary Material for "Deep Neural Network for Functional Graphical Models Structure Learning"

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In this Supplementary Material, we first provide the technical assumptions and definitions in Section S.1. We then provide the proofs of technical lemmas and Theorems 3.1–3.4 in Section S.2. In Section S.3, we demonstrate the details for Algorithm 3. Specific model examples showing that the other models are encompassed in our function class are provided in Section S.4. A graphical representation of the proposed model and additional simulation results for phase transition phenomenon and computational time are presented in Section S.5, and summary of notations used in Section 2 are given in Section S.6.

## S.1 Additional Assumptions and definition

**Assumption 3.** *There exists a DNN class $\mathcal{G}$, with structure parameter $L$ and $\{d_v\}_{v=2,\ldots,L}$ defined by Equation (2.2), satisfies that*

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**(a)** $L \geq \sup_{i=1,\ldots,p} J^i + 216 \sum_{u=1}^{J^i} (\lceil \gamma_u^i \rceil^2 + 1)$;

**(b)** $\min_{2\leq v\leq L} d_v \geq \sup_{i=1,\ldots,p} 81 m_i \max_{1\leq u\leq J^i} r_u^i (\lceil \gamma_u^i \rceil + s_u^i + 2)^{s_u^i+1} 3^{s_u^i+1}$;

**(c)** $\sup_{i=1,\ldots,p} \|\boldsymbol{m}^{(-i)}\|_1 \lesssim L \max_{2\leq v\leq L} d_v \asymp L \min_{2\leq v\leq L} d_v \asymp n^{\frac{\bar{s}}{2(2\bar{\gamma}+\bar{s})}}$, *with* $\bar{\gamma} = \tilde{\gamma}_{\bar{u}^i}^{\bar{i}}$, $\bar{s} = s_{\bar{u}^i}^{\bar{i}}$, *where* $(\bar{i}, \bar{u}_i) = \arg\min_{\substack{1\leq i\leq p \\ 1\leq u\leq J^i}} \frac{\tilde{\gamma}_u^i}{s_u^i}$;

**(d)** $\sup_{i=1,\ldots,p} m_i \lesssim \log n$.

**Assumption 4.** *For any node $i \in \{1, \ldots, p\}$, the DNN class $\mathcal{G}^i$ satisfies that*

**(a)** $L^i \geq J^i + 216 \sum_{u=1}^{J^i} (\lceil \gamma_u^i \rceil^2 + 1)$;

**(b)** $\min_{2\leq v\leq L^i} d_v^i \geq 81 m_i \max_{1\leq u\leq J^i} r_u^i (\lceil \gamma_u^i \rceil + s_u^i + 2)^{s_u^i+1} 3^{s_u^i+1}$;

**(c)** $\|\boldsymbol{m}^{(-i)}\|_1 \lesssim L^i \max_{2\leq v\leq L^i} d_v^i \asymp L^i \min_{2\leq v\leq L^i} d_v^i \asymp n^{\frac{\tilde{\gamma}^i}{2(2\tilde{\gamma}^i+\tilde{s}^i)}}$, *if* $\underline{N}_i \geq N_i^*$,

$\|\boldsymbol{m}^{(-i)}\|_1 \lesssim L^i \max_{2\leq v\leq L^i} d_v^i \asymp L^i \min_{2\leq v\leq L^i} d_v^i \asymp (\underline{N}_i)^{\frac{\nu \tilde{s}^i}{2\tilde{\gamma}^i}}$, *if* $\underline{N}_i < N_i^*$,

*where* $\underline{N}_i = \min_{j\neq i} N_j$, $N_i^* = \left( n^{\frac{\tilde{\gamma}^i}{2\tilde{\gamma}^i+\tilde{s}^i}} / \log^9 n \right)^{1/\nu}$;

**(d)** $m_i \lesssim \log n$, *if* $\underline{N}_i \geq N_i^*$, *and* $m_i \lesssim \log \underline{N}^i$, *if* $\underline{N}_i < N_i^*$.

**Definition 1.** *A function $e : \mathbb{R}^s \to \mathbb{R}$ is called $\gamma$-Hölder smooth function if there exists a constant $c$, such that*

$$\sum_{\boldsymbol{\alpha}:|\boldsymbol{\alpha}|<\gamma} \|\partial^{\boldsymbol{\alpha}} e\|_\infty + \sum_{\boldsymbol{\alpha}:|\boldsymbol{\alpha}|=\lfloor\gamma\rfloor} \sup_{\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^s, \boldsymbol{x}\neq\boldsymbol{y}} \frac{|\partial^{\boldsymbol{\alpha}} f(\boldsymbol{x}) - \partial^{\boldsymbol{\alpha}} f(\boldsymbol{y})|}{\|\boldsymbol{x}-\boldsymbol{y}\|_\infty^{\gamma-\lfloor\gamma\rfloor}} \leq c,$$

*where $\|\cdot\|_\infty$ is the uniform norm of a function.*

## S.2 Technical proofs

In this section, we provide technical lemmas and proofs that complement the main article. In Section S.2.1, we provide the proof of the equivalent condition of the conditional independence. In Section S.2.2, we provide technical proofs that lay the foundation for the main theorems. Firstly, Lemma S.2.1 establishes the approximation theory for the neural network, demonstrating that for any true function in $\mathcal{F}^i$, the approximation error of some DNN estimator within the considered function class $\mathcal{G}^i$ of the $i$-th node can be made arbitrarily small. Secondly, Lemma S.2.2 demonstrates the estimation error of the proposed DNN estimator in the statistical perspective, where the consistency holds when the DNN class has the optimal structure. Lastly, we provides the Lemma S.2.3, which is the preliminary result for Lemma S.2.2. It provides the statistical upper bound of the difference between the approximation error and the measurement error caused by randomness. This bound is effectively controlled when the true function under the $i$-th node is well approximated within an appropriately selected DNN class.

### S.2.1 Proof of Lemma 2.1

*Proof.* By definition, $\boldsymbol{\xi}^i \perp\!\!\!\perp \boldsymbol{\xi}^j | \boldsymbol{\xi}^{-(i,j)}$ is equivalent to the factorization as

$$\pi(\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^p) = \pi\left(\boldsymbol{\xi}^i | \boldsymbol{\xi}^{-(i,j)}\right) \pi\left(\boldsymbol{\xi}^j | \boldsymbol{\xi}^{-(i,j)}\right) \pi\left(\boldsymbol{\xi}^{-(i,j)}\right),$$

yielding that

$$\log \pi(\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^p) = \log \pi\left(\boldsymbol{\xi}^i | \boldsymbol{\xi}^{-(i,j)}\right) + \log \pi\left(\boldsymbol{\xi}^j | \boldsymbol{\xi}^{-(i,j)}\right) + \log \pi\left(\boldsymbol{\xi}^{-(i,j)}\right).$$

If $\pi$ is continuously differentiable, it is obvious that

$$\partial_{\boldsymbol{\xi}^i} \partial_{\boldsymbol{\xi}^j} \log \pi(\boldsymbol{\xi}^1 \ldots, \boldsymbol{\xi}^p) = \mathbf{0}, \tag{S.2.1}$$

for any $\boldsymbol{\xi}$ (arbitrary $m_i$ later). Conversely, if a strictly positive continuously differentiable $\pi(\boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^p)$ satisfies Equation (S.2.1), then the conditional independence holds. The rest of proof is analogue to Lemma 4.1 in Spantini et al. (2018), so it is omitted. □

### S.2.2 Technical lemmas and proofs for the DNN estimator

We denote the two essential convergence rates for the $i$-th node by

$$A_n^i = \left(\left(L^i \min_{2 \le v \le L^i} d_v^i\right)^2 + L^i \min_{2 \le v \le L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right)^{1/2} n^{-1/2}$$
$$\times \log^{1/2}\left(L^i \left(\min_{2 \le v \le L^i} d_v^i\right)^2 + \min_{2 \le v \le L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right),$$

and

$$B_n^i = \left(\left(L^i \min_{2 \le v \le L^i} d_v^i\right) \log^{-1}\left(L^i\right) \log^{-1}\left(\min_{2 \le v \le L^i} d_v^i\right)\right)^{-\frac{2\tilde{\gamma}^i}{\tilde{s}^i}}.$$

The following lemma provides the approximation theory for the true $\boldsymbol{f}^i$ within $\mathcal{G}^i$

**Lemma S.2.1.** *(Approximation error) For any $\boldsymbol{f}^i \in \mathcal{F}^i$, there exists $\widetilde{\boldsymbol{f}}^i \in \mathcal{G}^i$, such that when $L^i \to \infty, \min_{2 \le v \le L^i} d_v^i \to \infty$, it follows that $\|\widetilde{f}_l^i - f_l^i\|_\infty \lesssim B_n^i$, for any $l = 1, \ldots, m_i$.*

The proof of Lemma S.2.1 is similar to the proof of Lemma A.18 in Liu et al. (2021) for any fixed $\boldsymbol{m}^{(-i)}$ and $m_i$, thus is omitted. Based on Lemma S.2.1, the following lemma provides the estimation error of the regression estimator $\boldsymbol{f}^i$.

**Lemma S.2.2.** *(Consistency of $\widehat{\boldsymbol{f}}^i$) Under Assumptions 1 and 2, when*

$$L^i \min_{2 \le v \le L^i} d_v^i \lesssim \sqrt{n}$$

*and $L^i \min_{2 \le v \le L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1 \lesssim n$, it suffices to show that*

$$\|\widehat{\boldsymbol{f}}^i - \boldsymbol{f}^i\|_{n,\infty} = O_p\left(A_n^i + B_n^i\right).$$

*Proof.* In the following, define the residual $\Delta_l = \xi_l^i - f_l^i(\boldsymbol{\xi}^i)$ and its sample realization as $\Delta_{kl}$. Throughout the proof, we assume that $L^i$ and $\min_{2\le v\le L^i} d_v^i$ are relatively large as $n$ goes to infinity, such that the approximation error $B_n^i \le \log n$. This can be easily achieved as according to Lemma S.2.1, $B_n^i$ can be arbitrarily small when the $\mathcal{G}^i$ is large enough. According Lemma S.2.1 and Assumption 1, it follows that $\sum_{l=1}^{m_i} \|\widetilde{f}_l^i - f_l^i\|_\infty \lesssim m_i B_n^i \lesssim \log n B_n^i$.

First, by definition, $\widehat{\boldsymbol{f}}^i$ satisfies $\sum_{k=1}^n \|\boldsymbol{\xi}_k^{(-i)} - \widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_2^2 \le \sum_{k=1}^n \|\boldsymbol{\xi}_k^{(-i)} - \widetilde{\boldsymbol{f}}^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_2^2$, which is equivalent to

$$\begin{aligned}
&\sum_{l=1}^{m_i} \|f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 + 2\sum_{l=1}^{m_i}\langle \Delta_l, f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\rangle_n \\
\le &\sum_{l=1}^{m_i} \|f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widetilde{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 + 2\sum_{l=1}^{m_i}\langle \Delta_l, f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widetilde{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\rangle_n \\
\le &m_i\left(B_n^i\right)^2 + 2\sum_{l=1}^{m_i}\langle \Delta_l, f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widetilde{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\rangle_n
\end{aligned}$$

by admitting the decomposition

$$n^{-1}\sum_{k=1}^n \|\boldsymbol{\xi}_k^{(-i)} - \widehat{\boldsymbol{f}}^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_2^2 = \sum_{l=1}^{m_i} \|\Delta_l + f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2$$

and

$$n^{-1}\sum_{k=1}^n \|\boldsymbol{\xi}_k^{(-i)} - \widetilde{\boldsymbol{f}}^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_2^2 = \sum_{l=1}^{m_i} \|\Delta_l + f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widetilde{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2.$$

Accordingly, we have

$$\begin{aligned}
&\sum_{l=1}^{m_i} \|f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \\
\le &m_i\left(B_n^i\right)^2 + 2\sum_{l=1}^{m_i}\langle \Delta_l, f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widetilde{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\rangle_n + 2\sum_{l=1}^{m_i}\langle \Delta_l, \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\rangle_n \\
\le &m_i\left(B_n^i\right)^2 + 2\sum_{l=1}^{m_i} \|\widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n \|\Delta_l\|_n + 2B_n^i\sum_{l=1}^{m_i}\|\Delta_l\|_n \\
\le &m_i\left(B_n^i\right)^2 + 2\sqrt{\sum_{l=1}^{m_i} \|\widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2}\sqrt{\sum_{l=1}^{m_i}\|\Delta_l\|_n^2} + 2B_n^i\sum_{l=1}^{m_i}\|\Delta_l\|_n.
\end{aligned}$$

The Chebyshev's inequality implies that

$$\begin{aligned}
\mathbb{P}\left(\|\Delta_l\|_n > \log n, l = 1,\ldots,m_i\right) &\le \sum_{l=1}^{m_i}\mathbb{P}\left(\|\Delta_l\|_n > \log n\right) \\
&\le \sum_{l=1}^{m_i}\mathrm{E}\left(\Delta_l^2\right)\log^{-2}(n) \lesssim \log^{-1} n.
\end{aligned}$$

Therefore, we conclude that with probability at least $1-\log^{-1} n$, it follows that $\|\Delta_l\|_n \leq \log n$ for any $l=1,\ldots,m_i$, which further implies that

$$\begin{aligned}
&\sum_{l=1}^{m_i} \| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \\
\leq & m_i\left(B_n^i\right)^2 + 2\sqrt{\sum_{l=1}^{m_i} \| \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2}\sqrt{m_i}\log n + 2B_n^i m_i \log n.
\end{aligned}$$

By manipulating the quadratic form, we have

$$\sqrt{\sum_{l=1}^{m_i} \| \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2} \leq 8m_i \log n.$$

We denote the aforementioned event as $\mathcal{H}^*$, with $\mathbb{P}\left(\mathcal{H}^*\right) > 1-\log^{-1} n$.

Next, by using Chebyshev's inequality, it follows that

$$\begin{aligned}
\mathbb{P}\left(|\langle \widetilde{f}_l^i - f_l^i, \Delta_l\rangle_n| > B_n^i \frac{\log n}{n}\right) &\leq \mathrm{E}\left(|\langle \widetilde{f}_l^i - f_l^i, \Delta_l\rangle_n^2\right)\left(B_n^i \frac{\log n}{n}\right)^{-2} \\
&\leq \left(B_n^i n^{-1}\right)^2 \sum_{l=1}^{n} \mathrm{E}\left(\Delta_l^2\right)\left(B_n^i \frac{\log n}{n}\right)^{-2} \\
&= \mathrm{E}\left(\Delta_l^2\right)\log^{-2} n,
\end{aligned}$$

which implies that under Assumption 1, $|\widetilde{f}_l^i - f_l^i, \Delta_l\rangle_n| \leq B_n^i \frac{\log n}{n}$ for all $l = 1,\ldots,m_i$, with the probability at least $1-\log^{-1} n \max_l \mathrm{E}\left(\Delta_l^2\right)$. We denote the aforementioned event as $\mathcal{H}^{**}$, with $\mathbb{P}\left(\mathcal{H}^{**}\right) > 1-\log^{-1} n$.

By Lemma S.2.3 and Assumption 1, we have $\mathbb{P}\left(\mathcal{H}_\zeta\right) > 1-\log^{-1} n$, where $\mathcal{H}_\zeta$ is the event of Equation (S.2.2) from Lemma S.2.3 for any desired $\zeta$. In the following, we take a integer $r \equiv \lceil \log_2\left(8m_i n \log n\right)\rceil$ and $\zeta' > n^{-1}$, such that $8m_i \log n \leq 2^r n^{-1} \leq 2^r \zeta' < n$, and define the event $\mathcal{H} = \mathcal{H}^* \bigcap \mathcal{H}^{**} \bigcap \left(\bigcap_{\ell=1}^r \mathcal{H}_{2^\ell \zeta'}\right)$.

We first show that the probability that $\mathcal{H}$ occurs with probability goes to 1 as $n$ goes to infinity. In fact, it follows directly that

$$\begin{aligned}
\mathbb{P}\left(\mathcal{H}\right) &\geq 1 - r\max \mathbb{P}\left(\mathcal{H}_{2^\ell\zeta'}^c\right) - \mathbb{P}\left((\mathcal{H}^*)^c\right) - \mathbb{P}\left((\mathcal{H}^{**})^c\right) \\
&\gtrsim 1 - rm_i \log^{-2} n - 2\log^{-1} n \\
&\gtrsim 1 - \frac{m_i \log m_i + m_i \log n}{\log^2 n} - \log^{-1} n \\
&= 1 - o(1),
\end{aligned}$$

where the equality is by Assumption 1.

Moreover, since $\| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \leq 2^r\zeta'$ for all $l = 1,\ldots,m_i$, there exists an $\ell'$, such that $2^{\ell'-1}\zeta' \leq \| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \leq 2^{\ell'}\zeta'$, which

automatically leads to

$$
\begin{aligned}
&\sum_{l=1}^{m_i} \| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \\
\leq & m_i 2^{\ell'} \zeta' C\zeta \sqrt{\frac{L^i\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)}{n}} \\
& \times \sqrt{\log^5(n)\log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)} \\
& + \exp\left(-9\kappa_2/\kappa_1 \exp\left(\kappa_1 c\right)\right) + 2m_i \log n B_n^i n^{-1/2} + \left(B_n^i\right)^2,
\end{aligned}
$$

by the derivation under $\mathcal{H}^*$ and Lemma S.2.3. Therefore, by choosing

$$
\begin{aligned}
&(20 m_i \log n)^{-1} > \zeta' \\
&\geq \max\Bigg\{ 8m_i C\zeta \sqrt{\frac{L^i\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)}{n}} \\
&\times \sqrt{\log^5(n)\log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)} \\
&+ \exp\left(-9\kappa_2/\kappa_1 \exp\left(\kappa_1 c\right)\right), 2\left(m_i n^{-1/2}\log n + B_n^i\right), n^{-1}\Bigg\},
\end{aligned}
$$

it follows that $\sum_{l=1}^{m_i} \| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \leq 2^{\ell'}\left(\zeta'\right)^2$. Therefore, it holds recursively and results in

$$
\| f_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right) - \widehat{f}_l^i\left(\boldsymbol{\xi}_k^{(-i)}\right)\|_n^2 \leq \zeta', \forall l = 1, \ldots, m_i, \textit{ under } \mathcal{H},
$$

where

$$
\begin{aligned}
\zeta' = 8m_i \max\Bigg\{ & C\zeta \sqrt{\frac{L^i\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)}{n}} \\
& \times \sqrt{\log^5(n)\log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1\right)} \\
& + C\zeta \frac{\log^2 n}{n}, \left(n^{-1/2}\log n + B_n^i\right), n^{-1}\Bigg\}.
\end{aligned}
$$

The proof is complete. □

**Definition 2.** *For $\kappa > 0$, a subset $\{f_k \in \mathcal{G}\}_{k\geq 1}$ is called a $\kappa$-covering set of $\mathcal{G}$ with respect to $\|f\|_n$, if for all $f \in \mathcal{G}$, there exists an $f_k$ such that $\|f_k - f\|_n \leq \kappa$. The $\kappa$-covering number of $\mathcal{G}$ with respect to $\|\cdot\|_n$ is defined by*

$$
\mathcal{N}(\kappa, \mathcal{G}, \|\cdot\|_n) = \inf\left\{ N \in \mathbb{N} : \exists f_1, \ldots, f_N, s.t. \mathcal{G} \subset \bigcup_{k=1}^{N} \{f \in \mathcal{G} : \|f_k - f\|_n \leq \kappa\} \right\}.
$$

**Definition 3.** *For a function class $\mathcal{G}$, whose element is defined on $\mathcal{D}$, we say $\mathcal{G}$ has pseudo-dimension $d$, denoted by PD $(\mathcal{G}) \equiv d$, if $d$ is the maximum cardinality of a subset of $\mathcal{D}$ that is pseudo-shattered (Definition 11.1 in Anthony and Bartlett (2009)) by $\mathcal{G}$.*

Equipped with Definitions 2 and 3, we introduce Lemma S.2.3, which is a critical component of Lemma S.2.2.

**Lemma S.2.3.** *(Estimation error) Consider arbitrary function $\boldsymbol{f} \in \mathcal{G}^i$. Under Assumptions 1 and 2, with probability at least $1 - m_i \log^{-2} n$, for any $l \in 1, \ldots, m_i$, when $\zeta \in (n^{-1}, n)$, it follows that*

$$\begin{aligned}
&\sup_{\|f_l - f_l^i\|_n \le \zeta} \langle f_l - f_l^i, \Delta_l \rangle_n \\
&\le C\zeta \sqrt{\frac{L^i \left( (L^i \min_{2 \le v \le L^i} d_v^i)^2 + L^i \min_{2 \le v \le L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1 \right)}{n}} \\
&\times \sqrt{\log^5(n) \log \left( L^i \left( \min_{2 \le v \le L^i} d_v^i \right)^2 + L^i \min_{2 \le v \le L^i} d_v^i \|\boldsymbol{m}^{-i}\|_1 \right)} \\
&+ C\zeta \frac{\log^2(n)}{n},
\end{aligned} \tag{S.2.2}$$

*where $C$ is a universal constant.*

*Proof.* First, we partition $\Delta_l$ to two parts: $\Delta_l^{(1)} = \Delta_l \mathbb{I}\left(|\Delta_l| > 6 \log n / \kappa_1\right)$ and $\Delta_l^{(2)} > \Delta_l \mathbb{I}\left(|\Delta_l| \le 6 \log n / \kappa_1\right)$, and define the corresponding components

$$Z^{(1)} = \sup_{\|f_l - f_l^i\|_n \le \zeta} n^{-1} \sum_{k=1}^{n} \left(f_l - f_l^i\right) \left( \Delta_l^{(1)} - \mathrm{E}\left( \Delta_l^{(1)} | \boldsymbol{\xi}^{(-i)} \right) \right)$$

and

$$Z^{(2)} = \sup_{\|f_l - f_l^i\|_n \le \zeta} n^{-1} \sum_{k=1}^{n} \left(f_l - f_l^i\right) \left( \Delta_l^{(2)} - \mathrm{E}\left( \Delta_l^{(2)} | \boldsymbol{\xi}^{(-i)} \right) \right).$$

As a result, the desired bound can be achieved by $\sup_{\|f_l - f_l^i\|_n \le \zeta} \langle f_l - f_l^i, \Delta_l \rangle_n \le Z^{(1)} + Z^{(2)}$.

To bound $Z^{(1)}$, it follows that

$$\begin{aligned}
\mathrm{E}\, Z^{(1)} &\le \zeta\, \mathrm{E} \left\{ \left( n^{-1} \sum_{k=1}^{n} \left( \Delta_l^{(1)} - \mathrm{E}\left( \Delta_l^{(1)} | \boldsymbol{\xi}^{(-i)} \right)^2 \right) \right)^{1/2} \right\} \\
&\le \zeta \left( n^{-1} \sum_{k=1}^{n} \mathrm{E} \left( \Delta_l^{(1)} - \mathrm{E}\left( \Delta_l^{(1)} | \boldsymbol{\xi}^{(-i)} \right)^2 \right) \right)^{1/2} \\
&\le \zeta \left( n^{-1} \sum_{k=1}^{n} \mathrm{E} \left( \left( \Delta_l^{(1)} \right)^2 | \boldsymbol{\xi}^{(-i)} \right) \right)^{1/2}
\end{aligned}$$

To bound $Z^{(2)}$, we need the results from the following two steps.

*Step 1.* For any $l = 1, \ldots, m_i$, define the considered function space $\mathcal{S}_l(\zeta) \equiv \{f_l - f_l^i : \|f_l - f_l^i\|_n \leq \zeta\}$, and define the $S_l^q$ be some $2^{-q}\zeta$-covering of $\mathcal{S}_l(\zeta)$, with respect to $\|\cdot\|_n$. Let $h_l^q(f_l) \in S_l^q$ be the function within the $2^{-q}\zeta$ neighborhood of any $\widetilde{f}_l \in \mathcal{S}_l(\zeta)$. By decomposing $\widetilde{f}_l = \widetilde{f}_l - h_l^J(\widetilde{f}_l) + \sum_{q=1}^J \left(h_l^J(\widetilde{f}_l) - h_l^J(\widetilde{f}_{l-1})\right)$, it follows that

$$\begin{aligned}\langle \widetilde{f}_l, \Delta_l\rangle_n &\leq \langle \widetilde{f}_l - h_l^J(\widetilde{f}_l), \Delta_l\rangle_n + \sum_{q=1}^J \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n \\ &\leq 2^{-J}\|\Delta_l\|_n + \sum_{q=1}^J \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n.\end{aligned}$$

*Step 2.* For each $q$, consider the class

$$T_q = \left\{\widetilde{h}^q - \widetilde{h}^{q-1} : \|\widetilde{h}^q - \widetilde{h}^{q-1}\|_n \leq 3 \times 2^{-q}\zeta, \widetilde{h}^q \in S_l^q, \widetilde{h}^{q-1} \in S_l^{q-1}\right\},$$

as it follows that

$$\|h_l^q(\widetilde{f}_l) - \widetilde{f}_l\|_n \leq \|\widetilde{f}_l - h_l^{q-1}(\widetilde{f}_l)\|_n + \|h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l)\|_n \leq 3 \times 2^{-q}\zeta.$$

Then we have

$$\begin{aligned}&\exp\left(\left(\frac{\kappa_1 n \log(|T_q|)}{3\zeta \log n}\right)^{1/2} \mathrm{E}\left(\sup_{h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l) \in T_q} \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n\right)\right) \\ \leq &\mathrm{E}\left(\exp\left(\sup_{h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l) \in T_q} \left(\frac{\kappa_1 n \log(|T_q|)}{3\zeta \log n}\right)^{1/2} \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n\right)\right) \\ \leq &\sum_{\sup_{h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l) \in T_q}} \mathrm{E}\left(\exp\left(\left(\frac{\kappa_1 n \log(|T_q|)}{3\zeta \log n}\right)^{1/2} \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n\right)\right) \\ \leq &|T_q| \exp\left(\frac{\zeta^2}{2^{2q+1} n} \times \frac{\kappa_1 n \log(|T_q|)}{3\zeta \log n} \times \frac{36 \log^2 n}{\kappa_1^2}\right),\end{aligned}$$

which further implies that

$$\begin{aligned}&\mathrm{E}\left(\sup_{h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l) \in T_q} \langle h_l^q(\widetilde{f}_l) - h_l^{q-1}(\widetilde{f}_l), \Delta_l\rangle_n\right) \\ \leq &\frac{216 \log^2 n}{2^q \kappa_1^2} \zeta \left(\frac{\log(|T_q|)}{n}\right)^{1/2} \\ \leq &\frac{512 \log^2 n}{\kappa_1^2} \int_{2^{-q-1}\zeta}^{2^{-q}\zeta} \left(\frac{\log \mathcal{N}(x, \mathcal{S}_l(\zeta), \|\cdot\|_n)}{n}\right)^{1/2} dx.\end{aligned}$$

Together with the result in *Step 1*, by choosing some $\delta$, such that $2^{-J-2}\zeta \leq \delta < 2^{-J-1}\zeta$, it follows that

$$\mathrm{E}\, Z^{(2)} \leq 4\delta \left(\mathrm{E}\left(\left(\Delta_l^{(2)}\right)^2 |\boldsymbol{\xi}^{(-i)}\right)\right)^{1/2} + \frac{512 \log^2 n}{\kappa_1^2} \int_\delta^\zeta \left(\frac{\log \mathcal{N}(x, \mathcal{S}_l(\zeta), \|\cdot\|_n)}{n}\right)^{1/2} dx.$$

*Step 3.* Following the Theorems 12.2 and 14.1 in Anthony and Bartlett (2009), it holds that

$$\begin{aligned}\log \mathcal{N}\left(x, \mathcal{S}_l(\zeta), \|\cdot\|_n\right) \leq & \mathrm{PD} \log\left(2enx^{-1} \sup_{k=1,\ldots,n} |f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right) - f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right)|\right) \\ \leq & C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) \\ & \times \log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) \\ & \times \log\left(2enx^{-1} \sup_{k=1,\ldots,n} |f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right) - f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right)|\right),\end{aligned}$$

where $C'$ is some universal constant. Therefore, since $\sup_{k=1,\ldots,n} |f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right) - f_l\left(\boldsymbol{\zeta}_k^{(-i)}\right)| \leq n\zeta < n^2$, for any $x > 0$, it holds that

$$\begin{aligned}\log \mathcal{N}\left(x, \mathcal{S}_l(\zeta), \|\cdot\|_n\right) \leq & C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) \\ & \times \log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) \log\left(2en^3x^{-1}\right).\end{aligned}$$

*Step 4.* By selecting $\delta = \zeta\left(C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) n^{-1}\right)^{1/2}$ and $1 \leq C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right)$, together with the results from *Step 2* and *Step 3*, we have

$$\begin{aligned}\mathrm{E}\, Z^{(2)} \leq & 4\zeta\left(C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) n^{-1}\right)^{1/2} \\ & \times \left(\mathrm{E}\left(\left(\Delta_l^{(2)}\right)^2 | \boldsymbol{\xi}^{(-i)}\right)\right)^{1/2} \\ & + \frac{512 \log^2 n}{\kappa_1^2} \zeta\left(C'\left(\left(L^i \min_{2\leq v\leq L^i} d_v^i\right)^2 + L^i \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right)\right. \\ & \left.\times \log\left(L^i\left(\min_{2\leq v\leq L^i} d_v^i\right)^2 + \min_{2\leq v\leq L^i} d_v^i \|\boldsymbol{m}^{(-i)}\|_1\right) n^{-1} \log\left(2en^5\right)\right)^{1/2},\end{aligned}$$

which completes the bound for $\mathrm{E}\, Z^{(2)}$.

Finally, by applying the Chebyshev's inequality, For any $x > 0$, it follows that

$\mathbb{P}\left(\sup_{\|f_l-f_l^i\|_n\le\zeta}\langle f_l-f_l^i,\Delta_l\rangle_n>x\right)\le x^{-1}\left(\mathrm{E}\,Z^{(1)}+\mathrm{E}\,Z^{(2)}\right)$. By letting

$$\begin{aligned}x=&C\zeta\sqrt{\frac{L^i\left(\left(L^i\min_{2\le v\le L^i}d_v^i\right)^2+L^i\min_{2\le v\le L^i}d_v^i\|\boldsymbol{m}^{-i}\|_1\right)}{n}}\\&\times\sqrt{\log^5(n)\log\left(L^i\left(\min_{2\le v\le L^i}d_v^i\right)^2+L^i\min_{2\le v\le L^i}d_v^i\|\boldsymbol{m}^{-i}\|_1\right)}\\&+C\zeta\frac{\log^2(n)}{n}\end{aligned}$$

and $C=516\kappa_1^{-2}$, the proof is complete.

□

## S.2.3 Proof of Theorems

### S.2.3.1 Proof of Theorem 3.1

*Proof.* Under Assumption 2, together with Lemma S.2.1 and S.2.2, the result follows. □

### S.2.3.2 Proof of Theorem 3.2

For the $\bar{i}$-th node with true function class $\mathcal{F}^{\bar{i}}$, the result directly follows by Theorem 3.1. For any $i\neq\bar{i}$, since

$$\sup_{i=1,\dots,p}\|\boldsymbol{m}^{(-i)}\|_1\lesssim L\max_{2\le v\le L}d_v\asymp L\min_{2\le v\le L}d_v\asymp n^{\frac{\bar{s}}{2(2\bar{\gamma}+\bar{s})}},$$

we have

$$L\max_{2\le v\le L}d_v\asymp L\min_{2\le v\le L}d_v\lesssim n^{1/2}$$

and

$$\sup_{i=1,\dots,p}\|\boldsymbol{m}^{(-i)}\|_1L\max_{2\le v\le L}d_v\asymp\sup_{i=1,\dots,p}\|\boldsymbol{m}^{(-i)}\|_1L\min_{2\le v\le L}d_v\lesssim n.$$

As a consequence, the result follows by Lemma S.2.1 and Lemma S.2.2, which completes the proof.

### S.2.3.3 Proof of Theorem 3.3

*Proof.* For $\underline{N}_i\ge N_i^*$, the proof directly follows the proof of Theorem 3.1. For $\underline{N}_i<N_i^*$, by Assumption 4, we have

$$\left(L^i\max_{2\le v\le L^i}d_v^i\right)^{-\frac{2\tilde{\gamma}^i}{\tilde{s}^i}}\asymp\left(L^i\min_{2\le v\le L^i}d_v^i\right)^{-\frac{2\tilde{\gamma}^i}{\tilde{s}^i}}\asymp\underline{N}_i^{-v}.$$

Therefore, the approximation rate in Lemma S.2.1 is derived. Additionally, we have $n^{-1}<\underline{N}_i^{-v\frac{2\tilde{\gamma}^i}{\tilde{s}^i}}$, and $L^i\max_{2\le v\le L^i}d_v^i\asymp L^i\min_{2\le v\le L^i}d_v^i\asymp\underline{N}_i^{\frac{v\tilde{s}^i}{2\tilde{\gamma}^i}}$, which

implies that with high probability,

$$\sup_{\boldsymbol{f}^i \in \mathcal{F}^i} \| \widehat{\boldsymbol{f}}^i \left( \widehat{\boldsymbol{\xi}}^{(-i)} \right) - \boldsymbol{f}^i \left( \widehat{\boldsymbol{\xi}}^{-i} \right) \|_{n,\infty} \lesssim \underline{N}_i^{-v},$$

by Lemma S.2.1 and Lemma S.2.2. Since any elements in $\mathcal{F}^i$ are Lipschitz continuous, we have

$$\sup_{\boldsymbol{f}^i \in \mathcal{F}^i} \| \boldsymbol{f}^i \left( \widehat{\boldsymbol{\xi}}^{(-i)} \right) - \boldsymbol{f}^i \left( \boldsymbol{\xi}^{-i} \right) \|_{n,\infty} \lesssim \sup_{j \neq i} \| \boldsymbol{\xi}^j - \widehat{\boldsymbol{\xi}}^j \|_\infty = O_p \left( \underline{N}_i^{-v} \right),$$

and the proof is complete. □

#### S.2.3.4 Proof of Theorem 3.4

*Proof.* The proof can be derived in the same fashion by the Proposition 1 in Lemhadri et al. (2021), where we consider the group version with group hierarchical proximal operator. For the $i$-th node detection, by fixing the finite number of input $\|\boldsymbol{m}^{(-i)}\|_1$ and output $m^i$, the result directly follows, thus is omitted. □

## S.3 Algorithm and Illustration

**Algorithm 3** Updating $\boldsymbol{B}_j^i$ and $\boldsymbol{W}_1^{(j)}$ in Model (2.4)

**Inputs:**
hierarchy coefficient $M$, learning rate $\alpha$, network weights $\boldsymbol{B}_j^i$ and $\boldsymbol{W}_j$.
Sort vec $\left( \boldsymbol{W}_1^{(j)} \right) \in \mathbb{R}^{\nu_j}$ and denote by $|\omega_{j(1)}| \geq \ldots \geq |\omega_{j(\nu_j)}|$, where $\nu_j = m_j d_2$
**for** $u = 0, \ldots, \nu_j$ **do**
    Calculate $\zeta_{ju} = \frac{M}{1+M^2} S_{\alpha\lambda} \left( \|\boldsymbol{B}_j^i\|_F + C \sum_{v=1}^{u} |\omega_{j(v)}| \right)$, $S_{\alpha\lambda}(z) = \text{sign}(z) \max\left(|z| - \alpha\lambda, 0\right)$.
**end for**
Select $u_j^*$ as the smallest integer from $\left\{ u : |\omega_{j(u)}| \geq \zeta_{ju} \geq |\omega_{j(u+1)}| \right\}$, where $\omega_{j(0)} = +\infty$, $\omega_{j(\nu_j+1)} = 0$.
Update $\boldsymbol{B}_j^i \leftarrow \frac{\zeta_{ju_j^*} \boldsymbol{B}_j^i}{M \|\boldsymbol{B}_j^i\|_F}$.
**for** $k = 1, \ldots, m_j$ **do**
    **for** $l = 1, \ldots, d_2$ **do**
        Update $W_{1kl}^{(j)} \leftarrow \text{sign}\left( W_{1kl}^{(j)} \right) \left( \zeta_{ju_j^*} \wedge \left| W_{1kl}^{(j)} \right| \right)$
    **end for**
**end for**
**Outputs:** $\boldsymbol{B}_j^i$ and $\boldsymbol{W}_1^{(j)}$

Figure S.1 presents a graphical illustration of the proposed `fgDNN` for neighborhood selection of node $i$, introduced in Section 2.

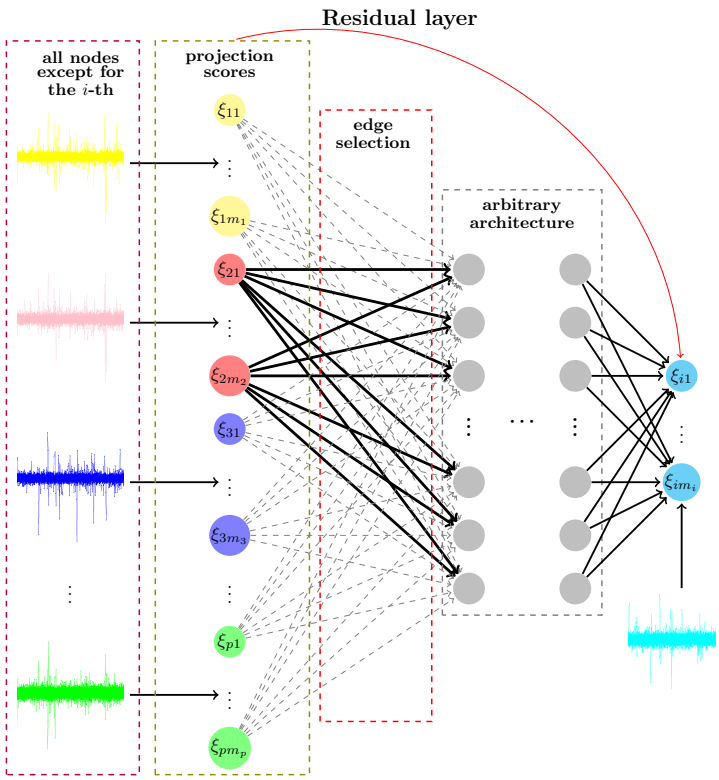


Figure S.1: Visualization of the proposed `fgDNN` for the $i$-th node neighborhood selection.

## S.4 Examples

***Gaussian graphical model in Zhao et al. (2024)*** Denote the precision matrix of $\left(\boldsymbol{\xi}^i, \boldsymbol{\xi}^{(-i)}\right)$ as

$$\begin{pmatrix} \boldsymbol{\Sigma}^i & \boldsymbol{\Sigma}^{i,(-i)} \\ \boldsymbol{\Sigma}^{i,(-i)} & \boldsymbol{\Sigma}^{(-i)} \end{pmatrix},$$

and it is easy to derive that

$$\boldsymbol{f}^i\left(\boldsymbol{\xi}^i\right) = \mathrm{E}\left(\boldsymbol{\xi}^i\right) + \boldsymbol{\Sigma}^{i,(-i)}\left(\boldsymbol{\Sigma}^{(-i)}\right)^{-1}\left(\boldsymbol{\xi}^{(-i)} - \mathrm{E}\left(\boldsymbol{\xi}^{(-i)}\right)\right).$$

Therefore, the desired $\mathcal{F}^i$ has $J^i = 2$, $r^i = \left(\|\boldsymbol{m}^{(-i)}\|_1, \|\boldsymbol{m}^{(-i)}\|_1, m^i\right)^\intercal$, $\boldsymbol{s}^i = \left(1, \|\boldsymbol{m}^{(-i)}\|_1\right)^\intercal$, $\boldsymbol{\gamma}^i = (\infty, \infty)^\intercal$, and Theorem 3.1 shows that our proposed method can achieve the parametric convergence rate $n^{-1}$ up to some logarithmic factor.

***Additive nonparametric graphical model in (Solea and Dette (2022))*** Following the model of Equation 4.1 in Solea and Dette (2022), we write

$$\boldsymbol{f}^i\left(\boldsymbol{\xi}^i\right) = \sum_{j \neq i} \sum_{\ell=1,\ldots,m_j} r^i_{\ell j}(\xi^j_\ell),$$

where $r^i_{\ell j}$ are some $\gamma$-Hölders across all $i, j, \ell$. As a consequence, the desired function class has $J^i = 2$, $r^i = \left(\|\boldsymbol{m}^{(-i)}\|_1, \|\boldsymbol{m}^{(-i)}\|_1, m^i\right)^\intercal$, $\boldsymbol{s}^i =$

$\big(1, \|\boldsymbol{m}^{(-i)}\|_1\big)^\intercal$, $\boldsymbol{\gamma}^i = (\gamma, \infty)^\intercal$, and Theorem 3.1 shows that our proposed method can achieve the classical nonparametric convergence rate $n^{-\frac{\gamma}{2\gamma+1}}$ up to some logarithmic factor.

## S.5 Additional Simulation Results

In Table 1, we present the average AUC and its standard error for `fgDNN` across 50 replicates for different sampling frequencies, $m_i$, with $p = 30$. It is evident that, for all models, the AUC increases as the minimum number of observation points, $\underline{N}_i$, for each functional curve at each node increases. This clear trend in AUC values supports Theorem 3.3.

| Model | $\underline{N}_i = 10$ | $\underline{N}_i = 25$ | $\underline{N}_i = 50$ | $\underline{N}_i = 100$ |
|---|---|---|---|---|
| A | 0.951(0.014) | 0.976(0.010) | 0.986(0.005) | 0.993(0.003) |
| B | 0.949(0.014) | 0.959(0.014) | 0.976(0.009) | 0.985(0.005) |
| C | 0.949(0.016) | 0.953(0.015) | 0.954(0.014) | 0.955(0.012) |
| D | 0.828(0.021) | 0.836(0.023) | 0.954(0.018) | 0.954(0.016) |

Table 1: The average AUC for `fgDNN` across $50$ replicates for different sampling frequency $\underline{N}_i$ with $p = 30$; the standard errors of AUC are given in the parentheses.

In Table 2, we present the average AUC and its standard error across 50 replicates for different sample size $n$, with $p = 100$.

In Table 3, we present the average computational time in minutes and its standard error across 50 replicates for different $p = 30, 50, 100$, with $n = 100$. The "FPCA-$g_Y$,OR" and "FPCA-$g_Y$,AND" share the same computational time, denoted as "FPCA-$g_Y$". Similarly, the "FPCA-$g_X$,OR" and "FPCA-$g_X$,AND" share the same computational time, denoted as "FPCA-$g_X$".

## S.6 Notations

# References


Anthony, M., and Bartlett, P. (2009), *Neural Network Learning*, Cambridge: Cambridge University Press.

Lemhadri, I., Ruan, F., Abraham, L., and Tibshirani, R. (2021), "LassoNet: A Neural Network with Feature Sparsity," in *Proceedings of the 38th International Conference on Machine Learning*, vol. 130 of *Proceedings of Machine Learning Research*, pp. 6237–6246.

Liu, R., Shang, Z., and Cheng, G. (2021), "On Deep Instrumental Variables Estimation," *arXiv preprint arXiv:2004.14954*.

| Model | $n$ | fgDNN | FPCA-$g_Y$,AND | FPCA-$g_Y$,OR | FPCA-$g_X$,AND | FPCA-$g_X$,OR | FGLasso | PSKL | npFGM |
|---|---|---|---|---|---|---|---|---|---|
| A | 100 | **0.991** **(0.002)** | 0.985 (0.003) | 0.976 (0.004) | 0.984 (0.003) | 0.975 (0.004) | 0.947 (0.006) | 0.925 (0.007) | 0.952 (0.005) |
| | 200 | **1.000** **(0.000)** | 0.989 (0.003) | 0.983 (0.004) | 0.987 (0.003) | 0.981 (0.004) | 0.955 (0.005) | 0.938 (0.006) | 0.966 (0.004) |
| | 300 | **1.000** **(0.000)** | 0.991 (0.003) | 0.987 (0.003) | 0.992 (0.003) | 0.988 (0.004) | 0.958 (0.005) | 0.951 (0.005) | 0.972 (0.003) |
| B | 100 | **0.987** **(0.003)** | 0.873 (0.140) | 0.863 (0.128) | 0.875 (0.145) | 0.864 (0.129) | 0.566 (0.131) | 0.637 (0.138) | 0.937 (0.004) |
| | 200 | **0.996** **(0.001)** | 0.889 (0.139) | 0.879 (0.131) | 0.899 (0.141) | 0.877 (0.125) | 0.634 (0.136) | 0.649 (0.126) | 0.956 (0.004) |
| | 300 | **1.000** **(0.000)** | 0.896 (0.122) | 0.892 (0.115) | 0.907 (0.133) | 0.880 (0.111) | 0.661 (0.130) | 0.664 (0.119) | 0.968 (0.003) |
| C | 100 | **0.986** **(0.006)** | 0.541 (0.170) | 0.545 (0.170) | 0.546 (0.166) | 0.542 (0.160) | 0.536 (0.160) | 0.558 (0.144) | 0.507 (0.119) |
| | 200 | **0.994** **(0.002)** | 0.513 (0.140) | 0.505 (0.140) | 0.486 (0.130) | 0.489 (0.145) | 0.519 (0.161) | 0.559 (0.136) | 0.496 (0.131) |
| | 300 | **1.000** **(0.000)** | 0.513 (0.147) | 0.495 (0.162) | 0.490 (0.158) | 0.501 (0.158) | 0.526 (0.140) | 0.614 (0.119) | 0.499 (0.129) |
| D | 100 | **0.934** **(0.009)** | 0.497 (0.045) | 0.504 (0.060) | 0.504 (0.049) | 0.504 (0.066) | 0.500 (0.061) | N/A (N/A) | 0.504 (0.059) |
| | 200 | **0.957** **(0.005)** | 0.529 (0.040) | 0.495 (0.069) | 0.490 (0.044) | 0.500 (0.056) | 0.526 (0.031) | N/A (N/A) | 0.511 (0.053) |
| | 300 | **0.992** **(0.003)** | 0.497 (0.045) | 0.518 (0.052) | 0.514 (0.039) | 0.501 (0.056) | 0.533 (0.066) | N/A (N/A) | 0.502 (0.049) |

Table 2: The average AUC for fgDNN across $50$ replicates for different sample size $n$ with $p = 100$; the standard errors of AUC are given in the parentheses.


Solea, E., and Dette, H. (2022), "Nonparametric and high-dimensional functional graphical models," *Electronic Journal of Statistics*, 16, 6175–6231.

Spantini, A., Bigoni, D., and Marzouk, Y. (2018), "Inference via low-dimensional couplings," *Journal of Machine Learning Research*, 19, 1–71.

Zhao, B., Zhai, P. S., Wang, Y. S., and Kolar, M. (2024), "High-dimensional functional graphical model structure learning via neighborhood selection approach," *Electronic Journal of Statistics*, 18, 1042–1129.

| Model | $p$ | `fgDNN` | FPCA-$g_X$ | FGLasso | PSKL | FPCA-$g_Y$ | npFGM |
|---|---|---|---|---|---|---|---|
| A | 30 | 271.623<br>(2.883) | 0.373<br>(0.013) | 0.120<br>(0.028) | 1.909<br>(0.050) | 0.463<br>(0.015) | 0.164<br>(0.008) |
| | 50 | 542.927<br>(5.537) | 1.680<br>(0.357) | 1.765<br>(0.194) | 3.837<br>(0.442) | 2.002<br>(0.497) | 0.531<br>(0.014) |
| | 100 | 1529.314<br>(51.966) | 10.064<br>(1.499) | 2.339<br>(2.084) | 18.638<br>(1.259) | 12.511<br>(1.380) | 3.541<br>(0.030) |
| B | 30 | 265.956<br>(2.968) | 1.190<br>(0.271) | 1.291<br>(1.103) | 0.496<br>(0.079) | 1.453<br>(0.343) | 0.466<br>(0.097) |
| | 50 | 554.548<br>(4.689) | 2.347<br>(0.671) | 2.255<br>(0.744) | 0.966<br>(0.101) | 2.510<br>(0.784) | 1.486<br>(0.312) |
| | 100 | 1348.116<br>(43.094) | 11.540<br>(2.003) | 6.521<br>(1.066) | 2.228<br>(0.155) | 13.306<br>(2.985) | 18.338<br>(42.279) |
| C | 30 | 274.461<br>(3.087) | 0.351<br>(0.070) | 0.066<br>(0.103) | 0.657<br>(0.081) | 0.652<br>(0.093) | 1.090<br>(0.045) |
| | 50 | 558.887<br>(4.238) | 1.449<br>(0.211) | 0.095<br>(0.125) | 0.768<br>(0.036) | 4.985<br>(0.074) | 14.298<br>(0.475) |
| | 100 | 1296.833<br>(44.837) | 9.307<br>(1.571) | 1.459<br>(4.097) | 3.758<br>(0.224) | 11.680<br>(2.046) | 39.459<br>(0.539) |
| D | 30 | 272.009<br>(3.266) | 0.813<br>(0.135) | 0.019<br>(0.007) | 0.000<br>(0.000) | 1.236<br>(0.228) | 6.439<br>(0.629) |
| | 50 | 549.536<br>(5.632) | 1.867<br>(0.511) | 0.063<br>(0.170) | 0.000<br>(0.000) | 2.651<br>(0.647) | 18.431<br>(1.201) |
| | 100 | 1553.686<br>(50.142) | 10.298<br>(1.917) | 0.270<br>(1.030) | 0.000<br>(0.000) | 15.207<br>(2.910) | 47.443<br>(1.268) |

Table 3: The average computational time (in minutes) across $50$ replicates for different sample size $n$ with $p = 100$; the standard errors of AUC are given in the parentheses.

| | |
|---|---|
| $\|\cdot\|_1$ | $L_1$ norm |
| $\|\cdot\|_2$ | $L_2$ norm |
| $\|\cdot\|_F$ | Frobenius norm |
| $\|\cdot\|_\infty$ | infinity norm |
| $\boldsymbol{X}(t) = (X^1(t), \ldots, X^p(t))^\intercal$ | $p$-dimensional multivariate functional data |
| $G = (\mathbb{V}, \mathbb{E})$ | an undirected graph |
| $\mathbb{V} = \{1, \ldots, p\}$ | the set of vertices |
| $\mathbb{E}$ | the set of edges; a subset of $\{(i,j) \in \mathbb{V} \times \mathbb{V} : i \neq j\}$ |
| $\boldsymbol{X}^{-(i,j)}(t)$ | $\boldsymbol{X}(t)$ with its $i$-th and $j$-th components removed |
| $\boldsymbol{\xi}_k^i$ | unobservable random coefficients of $X_k^i$ with respect to the population basis $\phi_\ell$ |
| $\left\{\widehat{\phi}_1, \ldots, \widehat{\phi}_{m_j}\right\}$ | basis functions |
| $\widehat{\boldsymbol{\xi}}_k^i = (\widehat{\xi}_{k,1}^i, \widehat{\xi}_{k,2}^i, \ldots)$ | an estimator of $\boldsymbol{\xi}_k^i := (\xi_{k,1}^i, \xi_{k,2}^i, \ldots)$ |
| $\boldsymbol{\xi}^i := \boldsymbol{\xi}_{(m_i)}^i = (\xi_1^i, \ldots, \xi_{m_i}^i)^\intercal$ | the first $m_i$ functional projection scores in the Karhunen-Loève expansion of $X^i(t)$ |
| $\boldsymbol{\xi}^{(-i)}$ | $(\boldsymbol{\xi}^1 \ \ldots, \boldsymbol{\xi}^{i-1}, \boldsymbol{\xi}^{i+1}, \ldots, \boldsymbol{\xi}^p)^\intercal$ |
| $\boldsymbol{f}^i$ | $\mathbb{R}^{\|\boldsymbol{m}^{(-i)}\|_1} \to \mathbb{R}^{m_i}$ |
| $\boldsymbol{m} = (m_j)_{j=1,\ldots,p}^\intercal$ | the truncation vector of functional projection scores |
| $\boldsymbol{m}^{(-i)} = (m_j)_{j \neq i}^\intercal$ | the truncation vector excluding the $i$-th node of interest |
| $\sigma(x) = (x)_+$ | the ReLU activation function |
| $L^i \in \mathbb{N}^+$ | the number of hidden layers |
| $\boldsymbol{d}^i = (d_1^i, \ldots, d_{L+1}^i) \in \mathbb{N}^{L^i+1}$ | the node vector of the hidden layer |
| $\boldsymbol{B}_j^i$ | a $m_i \times m_j$ weight matrix for the residual layer |
| $\boldsymbol{\eta} = (\boldsymbol{m}, L, \boldsymbol{d}, \lambda)$ | the tuning parameter set; $L$ is the number of hidden layer;<br>$\boldsymbol{d}$ is the vector of number of neurons; $\lambda$ is the initial tuning parameter |

Table 4: List of notations introduced in Section 2